ISSUE 01

# QUANICLE
## THE NEWSLETTER

CHRIST (DEEMED TO BE UNIVERSITY)
BANGALORE | DELHI NCR | PUNE

## COORDINATOR'S NOTE

It is with great pleasure that we present the inaugural issue of QUANICLE, the newsletter of the Centre for Quantum Technologies and Complex Systems (CQTCS).

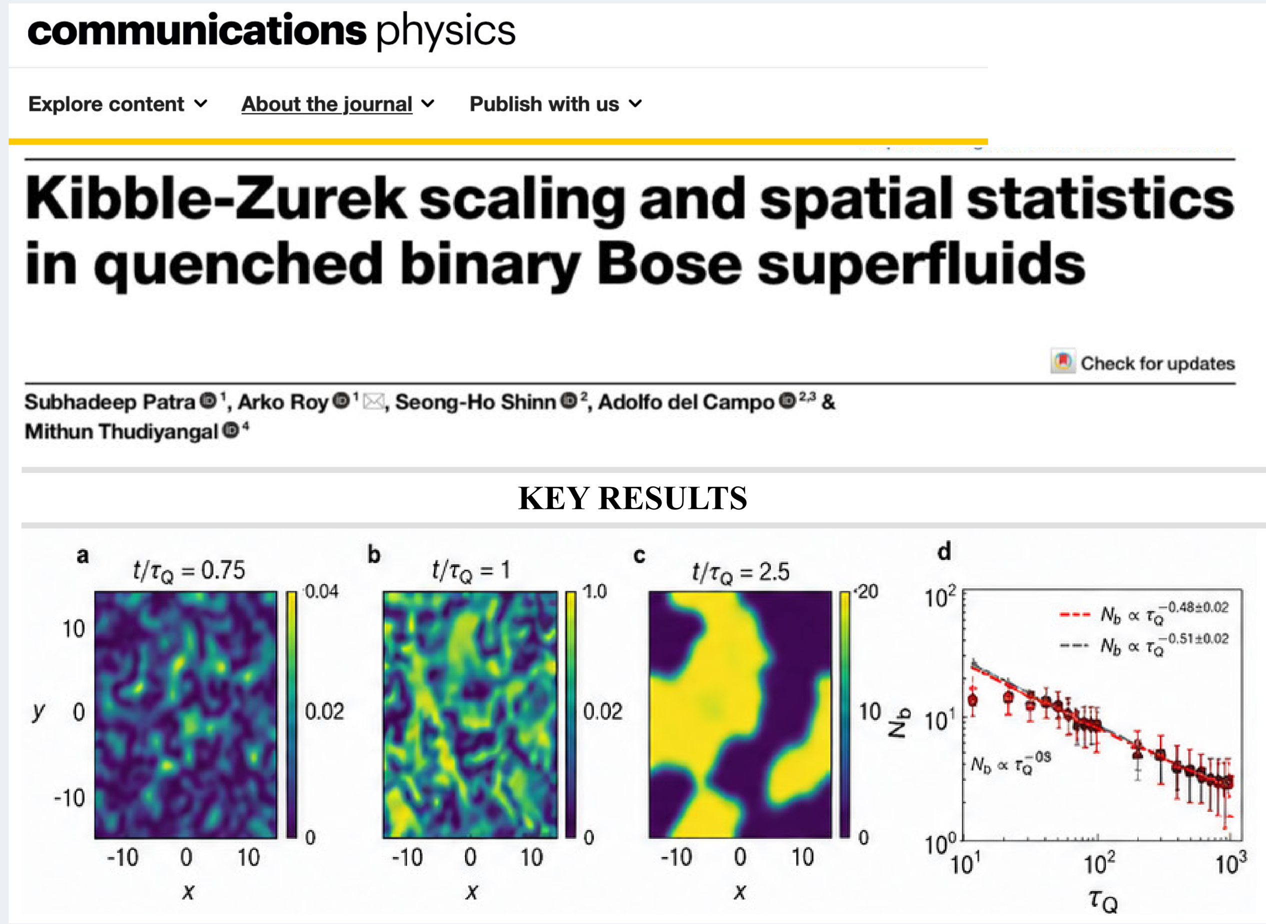
This platform will showcase our research excellence, celebrate achievements, and highlight opportunities for collaboration in the exciting world of quantum science and complex systems.

Together, let us continue to push the boundaries of knowledge and create a meaningful impact on society.

**Dr. Mithun Thudiyangal**
**Coordinator, CQTCS**

## RESEARCH SPOTLIGHT

### New Insights into Quantum Superfluid Dynamics


communications physics
Explore content ⌄ About the journal ⌄ Publish with us ⌄

**Kibble-Zurek scaling and spatial statistics in quenched binary Bose superfluids**

Check for updates

Subhadeep Patra[1], Arko Roy[1], Seong-Ho Shinn[2], Adolfo del Campo[2,3] & Mithun Thudiyangal[4]



Dr. Mithun Thudiyangal and collaborators from the University of Luxembourg have published their research article, "Kibble-Zurek Scaling and Spatial Statistics in Quenched Binary Bose Superfluids." in Communications Physics (Nature), highlighting significant advances in the study of non-equilibrium dynamics in quantum many-body systems.

The study explores universal features of far-from-equilibrium dynamics and introduces a framework for characterizing stochastic geometry in multicomponent quantum systems.

READ THE PAPER

## REPRESENTING THE CENTRE IN AUSTRIA

**Ms. Dipasha Chokeda**

**Mr. Sahil Satapathy**

12–15 July 2026 | Innsbruck, Austria

**H**ave been selected to participate in the Introductory Course on Ultracold Quantum Gases, jointly organized by Universität Innsbruck and the Institute for Quantum Optics and Quantum Information (IQOQI), Austria.

The program brings together young researchers from around the world to explore the fascinating physics of ultracold quantum gases.

## PUBLICATIONS

**Dr. Sreeja C S**

Dr. Sreeja C. S. published a research article in the Q1 journal IEEE Access, co-authored with Ph.D. scholar Mr. Vivin Krishnan, titled "Provably Adaptive Trust Dynamics in Context-Aware Zero-Trust Systems: A Formal Framework for Continuous Verification."

The paper introduces Zero-Trust Hybrid Adaptive Authentication (ZeTHAA), a continuous authentication and authorization framework integrating contextual attributes, authentication strength, behavioral evidence, and retry dynamics. ZeTHAA utilizes a probabilistic risk model and dual-policy thresholds to partition outcomes into allow, step-up, and block regions, enabling precise control over security-usability trade-offs


Journals & Magazines > IEEE Access > Volume: 14

**Provably Adaptive Trust Dynamics in Context-Aware Zero-Trust Systems: A Formal Framework for Continuous Verification**

Publisher: IEEE | Cite This | PDF

Vivin Krishnan ; C. S. Sreeja | All Authors

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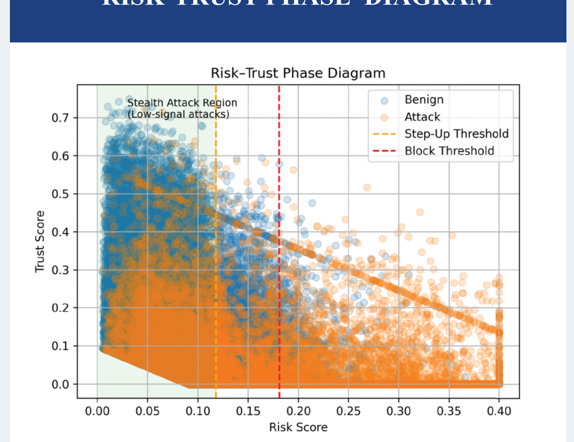
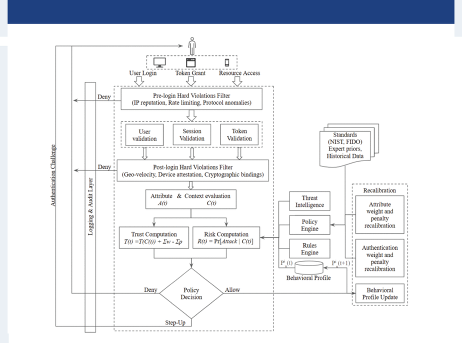
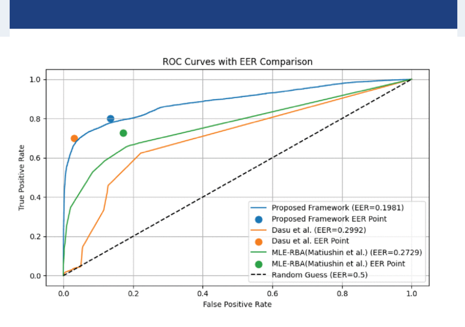
RISK-TRUST PHASE DIAGRAM



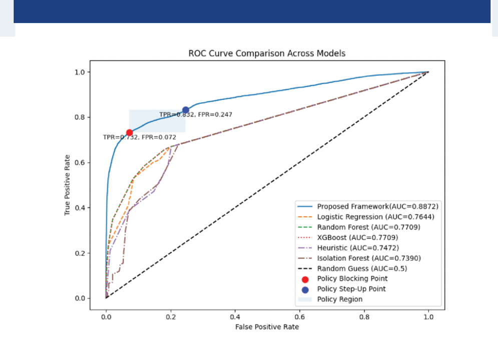

ARCHITECTURAL FRAMEWORK



ROC CURVE WITH EER POINTS



ROC CURVE COMPARISON



## MEET OUR GROUP LEADERS

**Dr. Mithun Thudiyangal**
Group Leader
**Research Focus**
Ultracold Quantum Matter, Quantum Algorithms & Quantum fluids

**Dr. Basudeb Sain**
Group Leader
**Research Focus**
Photonics / Quantum Photonics & Optical Metasurfaces

**Dr. Arindam Mallick**
Group Leader
**Research Focus**
Quantum Information, Quantum Simulation & Mathematical Physics

**Dr. Sreeja C S**
Group Leader
**Research Focus**
Quantum-Safe Cryptography & Cyber Security

**Dr. Ranjith Kumar K M**
Group Leader
**Research Focus**
Quantum Materials & Quantum Magnetism

## EVENT RECAP

Dr. Sreeja C S attended the Quantum Pitch Fest, held on 22–23 May 2026 at the Indian Institute of Science (IISc), Bengaluru, organized by WIN-CoE, IISc Bengaluru, IQTI, QuRP, and the Foundation for QC Innovation.

## UPCOMING EVENTS

**6 June** | Certificate Course: On Emerging Quantum Technologies | Online & Offline Modes

**" Exploring Quantum. Understanding Complexity. Engineering the Future. "**

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